Pandas

① Read data from csv file into a data frame.

```
import pandas as pd.
df1= pd.read_csv ('exmple.csv')
print (df)

df2= pd.read_excel(" Simple.xlsx")  // reading for excel file
print (df)

df= pd.read_json (" example.json")  // reading for json file
print (df3)
```

② Save data.

```
import pandas as pd
data = {
    "Name": ['Ram', 'shayam', "sid", Rohit']
    " Age": [30, 20, 19, 20]
    " City": ["agra', 'kanpur', 'Delhi', 'mumbia']

df = pd.DataFrame(data)
print(df)

df.to_csv (" output.csv", index=false)
df.to_excel (" output.exlsx," index=false)
df.to_json (" output.json", index=false)
```

③ ~~Read~~ look data

```
import pandas as pd.
df = pd.read_json (" simple.json")
print (df.head())   // first 5 row   (default)
print (df.head(10))  // first 10 row
print (df.tail())   // last 5 row   (default)
print (df.tail(10))  // last 10 row
```

④ Information about the data.

```
df= pd.read_json("simple.json")
print(df.info())
```

⑤ Description about the data.

```
df= pd.read_json("simple.json")
print(df.describe())
```

⑥ How big your data [shape ~~arr~~]

```
df= pd.read_json("Simple.json")
print(df.shape)
```

⑦ What are the name of Column

```
df= pd.read_json("Simple.json)
print(f'Column=: {df.columns}')
```

⑧ ~~find~~ select single column

```
df= pd.read_json("simple.json")
print(df['Name'])
```

⑨ Selecting multiple column

```
df= pd.read_json("Simple.json")
print(df[["Name", "Salary"]]
```

⑩ Filter from data fram

```
df = pd.read_json("simple.json")
high_salary= df[df['Salary']>5000]
print(high_salary)
```

```
               and
filtered= df[df['salary']>5000 & df['age']>30] // multiple
print(filtered)
                                or
filtered2= df[df['salary']>5000 | df['age']>30]
print(filtered2)
```

⑪ Add data in data frame.

(i)
```
df = pd.DataFrame(data)                    // new Column
df["Bonus"] = df['salary'] * 0.1
print(df)                                  // 10% of salary
```

(ii)
```
# using insert
# df.insert[loc, "Column Name", some data]
df = pd.DataFrame(data)
df.insert[0, "Employ ID", [10, 20, 30, 40]]
```

⑫ Updating value.

(i)
```
# df.loc[row_idx, "Column_name"] = new_value
df.loc[0, 'salary'] = 5500
```

(ii)
```
# increasing salary 5%          // update
df['Salary'] = df['Salary'] * 1.05
print(df)
```

⑬ Removing Column.
```
# df.drop(Columns=["Column Name"], inplace = True)
df.drop(columns=["performance"], inplace=True)
print(df)
```

⑭ Find missing value
```
df = pd.DataFrame(data)
missing = df.isnull()
print(missing)
print(df.isnull().sum())        // missing number count.
```

axis = 0 ⇒ rows
axis = 1 → Columns

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(15) Delet missing data (dropna())

```
df= pd.DataFrame(data)
df.dropna(inplace=True)
print(df)
```

(16) replace missing data (fillna())

```
df= pd.Dataframe(data)
df.~~drop~~ fillna(0, inplace=True)
print(df)
```

(17) Interpolation

```
data= {
"Time":[1,2,3,4,5],
"Val":[10,None,30,None,50]
}
df= pd.DataFrame(data)
print(f"Before Interpolation \n{df}")

df['val']= df['val'].interpolate(method="linear")
print(f"After Interpolation \n{df}")
```

(18) Sorting data.

(i) # Sorting data 1 Column sort_values()

```
# df.sort_value(by="ColumnName", True/False, inplace=True)
df= DataFrame(data)
df.sort_value(by="Age", assending=True, inplace=True)
print(df)
```

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(19) Groped and aggregation,

```
df= pd.DataFrame(data)
grouped= df.groupby("Age")["Salary"].sum()
print(grouped)
```

(20) marging and joining.

```
df_marged= pd.marge(df_coustomers, df_orders, on="Customer ID",
        how="outer")
print(df_marged)
```